

Optimization-Based Decision Support for Equitable Expansion of Federally Qualified Health Centers

Kim Erwin², Hemanshu Kaul¹, Alaittin Kirtışoğlu¹

Abstract

TODO

Keywords: Primary care access, Medically Underserved Areas, Health Equity, Federally Qualified Health Centers

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¹Department of Applied Mathematics, Illinois Institute of Technology, Chicago, IL 60616. E-mail: kaul@iit.edu, kirtisoglu@hawk.iit.edu

²Institute of Design, Illinois Institute of Technology, Chicago, IL 60616. E-mail: kerwin@id.iit.edu

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1 Introduction

The community health center program presents an opportunity to address critical public health vulnerabilities resulting from a lack of primary care services. The health center model has proven to increase access to primary care, reduce health disparities, lower costs, and improve community health. The number of clinic sites operated by these organizations has increased from less than 10,000 in 2015 to over 14,000 in 2021, serving 6 million additional patients (a 24% increase) over the same period [CITE: NACHC or HRSA annual report for clinic growth statistics].

In addition to providing direct services, health centers serve as the training ground for the next generation of primary care, behavioral health, and oral health providers [CITE: NACHC or HRSA workforce report]. A lack of access to primary care can weaken the public health response to future pandemics and could have dire consequences for patients, especially those facing compounding access barriers [8]. Patients with limited access to transportation, financial limitations, and other access barriers may forgo care rather than travel long distances to see a primary care provider.

Medically underserved refers to people who live in a federally designated Health Professional Shortage Area (HPSA) or Medically Underserved Area (MUA) or belong to a Medically Underserved Population (MUP). Since 2014, the number of medically disenfranchised people has nearly doubled from 56 million to over 100 million. A study by HealthLandscape at the American Academy of Family Physicians, commissioned by NACHC, found that over 100 million people in the United States may not have access to a usual source of primary care due to a shortage of providers in their community [9].

The Health Resources and Services Administration (HRSA) designates MUAs based on the Index of Medical Underservice (IMU), which incorporates four indicators: physician supply, poverty rate, infant mortality, and the proportion of elderly residents [5]. Areas with an IMU score of 62 or less qualify for MUA designation and become eligible for the establishment of Federally Qualified Health Centers (FQHCs). Nationally, \$3.7 billion in federal dollars were allocated for FQHCs in 2014 [9].

However, prior research has identified an important inconsistency: some geographic areas have IMU scores below the eligibility threshold but are not formally designated as MUAs [7]. This discrepancy creates a gap between measured healthcare need and formal designation, potentially affecting where federal resources are directed. MUA designation usually increases with poverty, but it actually declines once a neighborhood hits a certain tipping point of extreme poverty and racial segregation. Highly underserved areas often lack the community organization and resources required to successfully navigate the federal application process for MUA status [7].

Some very alarming statistics reveal population-level disparities within our social and health-care systems. One prominent example is the 30-year gap in life expectancy between people living in the poor, predominantly African American neighborhoods on Chicago’s South Side, as compared to those in the more affluent, predominantly white neighborhoods just nine miles away in Chicago’s Loop—the largest life expectancy gap in the United States. Real solutions are only possible with equitable strategies that recognize the contribution of historical and contemporary policies, economics, and health-care access [12].

In Chicago, 68 new clinics opened and only 12 closed since 2009, yet these additions often

failed to improve access in the areas that needed them most. Most new clinics were not located in the most severely underserved South and West Side neighborhoods, leading to a safety net that perpetuates racial and socioeconomic inequality [7]. Recent closures further strain the system: a Humboldt Park health clinic that served neighbors and low-income people for nearly a decade was closed [1], Howard Brown Health closed two clinics amid financial pressure [11], and Advocate Health Care closed nearly 50 clinics inside Walgreens stores [6].

The process for opening new FQHC facilities typically occurs through grant applications from local organizations rather than through systematic spatial planning. HRSA does not use formal optimization models like location-allocation or flow models [CITE: source for HRSA process description]. Community organizations apply for federal funding, and proposals are evaluated based on population need, access gaps, financial viability, and organizational capacity. As a result, facility placement may not fully reflect national or regional access optimization.

There is currently no large-scale optimization-based location-allocation model that integrates underserved area indicators, facility capacity constraints, population demand, and equity in geographic access. Most existing studies analyze healthcare access descriptively rather than optimizing facility placement. To the best of our knowledge, there is no single paper for Chicago providing actionable optimization-based strategies for decision makers.

We propose a flow-based spatial optimization framework for equitable planning of primary care access in Chicago. The framework:

- Computes IMU scores at finer geographic projections (census blocks) to identify the MUA designation gap at the boundaries,
- Proposes alternative IMU formulations such as probabilistic scores and smooth boundary transitions to address the sharp cutoff at $IMU = 62$,
- Designs a hierarchical location-allocation model with a referral system linking FQHCs to hospitals,
- Uses CTA public transit travel times reflecting real access barriers,
- Uses FalCom, an MCMC sampler for hierarchical capacitated facility location, as the computational engine,
- Provides an interactive decision-support dashboard for community organizations and investors preparing FQHC grant applications: given a proposed facility location, the dashboard runs a FalCom chain and returns a predicted HRSA application score, IMU score with smooth boundary version, patient flow distribution, referral network structure with HRSA compliance check, and ensemble robustness statistics across the distribution of feasible partitions.

2 Background

2.1 Federally Qualified Health Centers

Health centers need additional resources to sustain program growth and expand access to primary care in underserved communities. New FQHCs are usually not built through a centralized optimization plan. Instead, the process works roughly as follows: (i) an area is identified as underserved using metrics such as the IMU or HPSA scores; (ii) community organizations apply for federal funding to open or expand a health center; (iii) HRSA evaluates proposals based on population need, access gaps, financial viability, and organizational capacity; and (iv) HRSA awards Section 330 grants to selected applicants [5].

FQHCs primarily provide primary care, preventive care, and chronic disease management, but they do not provide most specialized or inpatient services. Patients must be referred to hospitals, specialists, and diagnostic facilities. HRSA requires formal referral relationships, and research shows that strong referral relationships are critical for community health center effectiveness [CITE: source for FQHC referral effectiveness].

2.2 MUA Designation and the IMU Score

An MUA is a geographical area designated as lacking access to primary care services and eligible for enhanced federal support. The geographical area can be a whole county, a group of neighboring counties, or a group of neighboring census tracts. HRSA is responsible for the evaluation of eligibility and designation of MUAs based on the IMU score [5].

The IMU is computed as a weighted sum of four indicators, each with a standardized contribution:

1. The ratio of primary care physicians per 1,000 residents,
2. The infant mortality rate,
3. The percentage of the population with incomes below the poverty level,
4. The percentage of the population aged 65 years or older.

$$\text{IMU} = S_{\text{phys}} + S_{\text{poverty}} + S_{65+} + S_{\text{IMR}} \quad (1)$$

$$S_{\text{phys}} = 28.7 \cdot \frac{P_{\text{max}} - P}{P_{\text{max}} - P_{\text{min}}} \in [0, 28.7] \quad (2)$$

$$S_{\text{poverty}} = 25.1 \cdot \frac{V_{\text{max}} - V}{V_{\text{max}} - V_{\text{min}}} \in [0, 25.1] \quad (3)$$

$$S_{65+} = 20.2 \cdot \frac{A_{\text{max}} - A}{A_{\text{max}} - A_{\text{min}}} \in [0, 20.2] \quad (4)$$

$$S_{\text{IMR}} = 26.0 \cdot \frac{M_{\text{max}} - M}{M_{\text{max}} - M_{\text{min}}} \in [0, 26.0] \quad (5)$$

The total IMU score ranges from 0 to 100, where lower values indicate greater underservice. Areas with $\text{IMU} \leq 62$ qualify for MUA designation.

For metropolitan areas, the rational service area is defined as a group of census tracts that represent a neighborhood of homogeneous socio-economic and demographic characteristics. Although DHHS shortage area designation methods take into account some non-spatial factors such as age and socio-economic status, they are primarily regional availability measures that quantify the distribution of supply versus demand within a predefined region, often expressed as a ratio of population to practitioner [10].

The advantage of such a regional availability approach is that it is simple, straightforward to implement (the data for physicians and population are readily available and boundaries can be easily located), and convenient to administer because the government infrastructure is already in place [10].

2.3 Limitations of the IMU and the Designation Gap

There is significant literature discussing the limitations of IMU and MUA designation. The IMU formula relies on weights created in the 1970s. Ricketts et al. [10] argue that the method no longer adequately captures modern healthcare access barriers. Healthcare utilization patterns and physician workforce distribution have changed substantially. Major determinants of healthcare access such as insurance coverage, transportation barriers, racial disparities, chronic disease prevalence are not included in the IMU formula.

The designation rule $\text{IMU} \leq 62$ creates a sharp cutoff: two areas with IMU scores of 61 and 63 receive entirely different policy treatment, yet their healthcare needs may be nearly identical. Kim et al. [7] show that some Chicago tracts meet IMU eligibility criteria but are not designated MUAs because of administrative boundaries. Census tracts that are eligible for MUA status but not designated were predominantly Black and poor, located on the South Side of Chicago.

We address these limitations in two ways. First, we compute IMU scores at the census tract level across Chicago to identify tracts that are eligible but undesignated. Second, we propose alternative scoring approaches such as probabilistic IMU scores and smooth boundary transition functions that replace the binary cutoff with a continuous measure of underservice (Section 3.3).

2.4 The Chicago Context

Kim et al. [7] used the spatial distribution of MUA designations and FQHCs in Chicago to evaluate differences between MUA-eligible neighborhoods that received designation and those that did not. Of Chicago’s census tracts, 391 were in affluent neighborhoods not eligible for MUA designation, mostly on the north, northwest side, and downtown. The 390 tracts that were eligible and designated for MUA status were clustered on the far north and west side. The 87 tracts that were eligible (IMU scores 48.6–61.7) but not designated as MUAs were predominantly on the South Side.

Neighborhoods with a greater number of community organizations per 1,000 residents were more likely to receive MUA designation, highlighting that administrative capacity, not just medical need drives designation outcomes [7].

Table 1: Distribution of FQHCs in Chicago by MUA status [7].

	Chicago	MUA	Non-MUA
Newly added clinics (since 2009)	68	41	3
FQHCs per 100,000 poor	39.6	40.8	15.5
Total number of FQHCs	187	112	10
Total population	2,683,422	1,252,783	259,226

2.5 Travel Time and Transit Equity

In the city of Chicago, 36.9% of African American households had no vehicle compared to 23.2% of White households [CITE: ACS or census vehicle ownership data]. Physical distance is therefore less important than public transit accessibility, as many underserved residents lack private vehicles.

Ermagun and Tilahun [3] study the equity of access to different destinations by public transit in Chicago. They find that the far south, southwest, and northwest of the city are among areas with lowest accessibility. Looking at accessibility to hospitals, the largest drops associated with population increases are among the elderly, low-income, and families with children aged 5–19. Despite the need of these cohorts to have access to healthcare, public transit does not serve them well, while providing better access to areas with higher percentages of highly-educated people. The South Side, which is predominantly African American and low-income, is among the least accessible areas. Chicago is ranked 3rd nationally for greatest accessibility by transit and 4th for greatest accessibility by walking [CITE: source for Chicago transit ranking].

In this work, we use CTA (Chicago Transit Authority) public transit travel times as the distance metric for our model, reflecting the actual transportation options available to the underserved population.

2.6 HRSA Compliance Requirements

HRSA requires that health center service sites be accessible to the patient population relative to where the population lives or works [CITE: HRSA Program Compliance Manual]. Health centers must consider distance and travel time between service sites when assessing impact on patient access. HRSA also requires that health centers establish and maintain collaborative relationships with other health care providers within the catchment area, local hospitals, and specialty providers to:

- Reduce non-urgent use of hospital emergency departments,
- Ensure continuity of care across community providers,
- Provide access to services not available through the health center.

Facility placement should therefore consider proximity to hospitals and referral networks. This motivates the hierarchical structure of our model, where each FQHC is assigned to a referral hospital.

3 Model and Formulation

3.1 Doubly-Constrained Spatial Flow Model

Let I denote the set of demand nodes (census tracts or blocks) and J the set of open FQHCs in a given spatial partition. Each demand node $i \in I$ carries demand d_i , representing realized patient counts from the UDS or census-based population. Each FQHC $j \in J$ has capacity c_j derived from UDS workforce and visit data. The travel cost t_{ij} is the CTA public transit travel time from i to j .

Distance decay. We model the friction of travel using an exponential decay function:

$$f(t_{ij}) = e^{-\lambda t_{ij}} \quad (6)$$

where $\lambda > 0$ is a decay parameter calibrated to observed patient flows via the UDS data.

Flow model. Patient flows x_{ij} from demand node i to FQHC j are given by:

$$x_{ij} = A_i \cdot B_j \cdot d_i \cdot c_j \cdot f(t_{ij}) \quad (7)$$

where A_i and B_j are balancing factors ensuring that both supply and demand constraints are simultaneously satisfied.

Balancing factors. The factors A_i and B_j are the unique solution to:

$$A_i = \frac{1}{\sum_{j \in J} B_j \cdot c_j \cdot f(t_{ij})} \quad (8)$$

$$B_j = \frac{1}{\sum_{i \in I} A_i \cdot d_i \cdot f(t_{ij})} \quad (9)$$

These are solved via Furness (IPF) iteration: initialize $B_j = 1$ for all j , then alternate updating A_i via (8) and B_j via (9) until convergence, typically within 20–50 iterations. The resulting flows satisfy:

$$\sum_{j \in J} x_{ij} = d_i \quad \forall i \in I \quad (10)$$

$$\sum_{i \in I} x_{ij} = c_j \quad \forall j \in J \quad (11)$$

Constraint (10) ensures all demand is allocated; constraint (11) ensures each FQHC operates at full capacity. A singly-constrained variant (demand side only, $\sum_i x_{ij} \leq c_j$) is also considered for scenarios where FQHCs may operate below capacity.

Role of FalCom. FalCom generates the spatial partition — determining which census tracts fall within each FQHC’s catchment area and which FQHCs are assigned to which referral hospitals. Given a partition, the flow model (7)–(9) is solved to compute realized flows x_{ij} . The energy function evaluated at each MCMC step is a flow-based equity or access score. A natural choice is the population-weighted average travel time:

$$E = \frac{\sum_{i \in I} \sum_{j \in J} x_{ij} \cdot t_{ij}}{\sum_{i \in I} d_i} \quad (12)$$

Alternative energy functions include the Gini coefficient of tract-level access $(\sum_j x_{ij} t_{ij})/d_i$ or the fraction of demand within a target travel time threshold. The FalCom sampler generates a distribution over partitions weighted by $\mu_\beta \propto \exp(-\beta E)$, concentrating on equitable, low-energy configurations while preserving sampling diversity.

Calibration. Because UDS provides both realized patient flows and facility capacity, λ in (6) can be estimated from observed data rather than chosen ad hoc. This is an empirical advantage over most facility location studies, which rely on synthetic demand or proxy travel costs.

3.2 Hierarchical Catchment Areas

In the context of Chicago, a hierarchical system divides facilities into levels: primary care (Level 1) and hospital/specialty care (Level 2). Tao et al. [14] propose a hierarchical 2SFCA (H2SFCA) method for measuring spatial accessibility to hierarchical facilities, considering varied catchment area sizes, distance decay effects, and transport modes at each level. Wan et al. [15] develop a three-step floating catchment area method that indicates strong potential for identifying health professional shortage areas.

3.3 Redesigning the IMU Score

The IMU formula is not merely outdated; it is structurally flawed in ways that systematically misallocate resources. In this section we identify the specific failures, review prior reform attempts and the political constraints that blocked them, and propose a redesigned scoring framework that integrates with the optimization model.

3.3.1 Why the Formula Is Wrong

Beyond the well-documented omission of insurance, transportation, and chronic disease burden [10], the IMU has deeper structural problems.

Miscalibrated conversion tables. Each indicator is mapped to a weighted score via lookup tables calibrated to 1970s national distributions. The national poverty rate, provider supply, infant mortality rate, and age distribution have all shifted substantially. A census tract that would have scored 10/25.1 on poverty in 1975 may score 18/25.1 today on the same

table, not because it worsened but because the scale no longer matches the distribution. The scores do not measure what they claim to measure.

Statistical noise in infant mortality. A typical census tract has 300–500 births per year. At this sample size, a single infant death changes the rate by 2–3 per thousand, which can shift the IMU by several points. The indicator carries enormous variance and almost no signal at the geographic level where IMU is computed. Funding decisions are made on noise.

Linear independence assumption. The four indicators are summed with fixed weights, implying that they are independent and equally substitutable. A tract with no providers but low poverty scores the same as a tract with adequate providers but extreme poverty. In reality these factors interact: poverty without providers is catastrophically worse than either alone. A linear sum cannot capture compounding disadvantage.

Scale compression at the extremes. The 0–25 mapping for each indicator flattens differences among the most underserved areas. A tract with zero providers and a tract with 5 per 100,000 may both map to 24 or 25. For optimization, this is fatal: the score must differentiate precisely where need is greatest. The bounded sum (0 to 100) saturates at both tails, preventing the score from reflecting how far below the threshold an area truly falls.

Provider counts ignore capacity. The physician-to-population ratio counts heads, not capacity. A tract with one physician seeing 500 patients and a tract with one physician seeing 3,000 patients receive the same ratio. Our provider matching reveals that 34 of 166 FQHC sites in Chicago have zero providers registered at their address in the NPPES, yet the IMU formula would still count those sites as serving their area.

3.3.2 Prior Reform Attempts

Ricketts et al. [10] proposed an alternative methodology, the Index of Primary Care Underservice, developed at the Cecil G. Sheps Center at UNC Chapel Hill. Their six-step method computes an age-sex adjusted effective population, counts clinician FTEs (including nurse practitioners and physician assistants, not only physicians), and applies a barrier adjustment derived from regression on nine indicators: poverty below 200% FPL, unemployment, racial minority share, Hispanic share, linguistic isolation, infant mortality or low birthweight, standardized mortality ratio, low population density, and share aged 65+. Each indicator is converted to a national percentile rank and then to a partial score via regression-derived weights. The final output is an adjusted population-to-clinician ratio.

HRSA adopted this methodology in a proposed rule published in the Federal Register on February 29, 2008 (73 FR 11232–11281) [4]. The proposal would have unified the separate MUA and HPSA designation systems into a single three-tier framework. HRSA received overwhelmingly negative public comments and withdrew the rule without finalizing it. Many areas that held MUA or HPSA designations would have lost them under the new formula, threatening Section 330 funding for existing FQHCs. A previous proposed rule in 1998 was similarly withdrawn.

Section 5602 of the Affordable Care Act (2010) required HRSA to establish a new methodology via negotiated rulemaking. A 28-member committee voted 21–2 in favor of a revised approach in October 2011, but the Negotiated Rulemaking Act required unanimity for HHS to publish the recommendation as an interim rule. HHS did not act. The 1975 formula remains in effect today [4].

3.3.3 The Political Constraint

Every reform attempt has failed for the same reason: the IMU score is fused to a binary funding gate. The chain is:

$$\text{IMU} \leq 62 \longrightarrow \text{MUA designation} \longrightarrow \text{Section 330 funding eligibility.}$$

A one-point difference between 62 and 63 means the difference between millions of dollars and zero. Any formula change that reshuffles scores threatens existing clinics serving real patients. The binary cutoff makes the formula politically unreformable: improving the measurement changes the money.

3.3.4 The Dilution Problem

If the city is partitioned into contiguous districts (as required for honest catchment area design), each district’s IMU is the population-weighted average of its census tracts. Contiguity forces each district to include neighboring tracts that may differ in need, pulling the average toward the middle. A district straddling a high-poverty tract and a moderate-income tract may average to an IMU of 63 or 64, just above the cutoff, even though half its population is severely underserved.

This creates a tension between honest boundaries and the binary threshold:

- Gerrymandering (the current system) allows applicants to draw boundaries that cherry-pick the worst tracts and push IMU below 62.
- Contiguous partitioning produces honest boundaries but dilutes concentrated pockets of need, causing more communities to fail the threshold.

The more honest the boundary-drawing process, the fewer districts qualify. Communities that genuinely need help are penalized for having slightly better neighbors. This is a structural failure of the binary cutoff, not of the partitioning method.

3.3.5 Proposed Framework: Continuous Composite Scoring

We propose replacing the binary IMU threshold with a continuous underservice score that serves three roles: ranking districts by need, functioning as the energy in the FalCom optimization, and allocating funding proportionally rather than through a binary gate.

Score definition. Let $\mathcal{X} = \{x_1, \dots, x_K\}$ denote a set of K indicators measured on each census tract t . The underservice score is

$$S(t) = \sum_{k=1}^K w_k \varphi_k(x_k(t)), \quad (13)$$

where $\varphi_k : \mathbb{R} \rightarrow [0, 1]$ is a monotone normalization function for indicator k , and $w_k > 0$ are weights satisfying $\sum_k w_k = 1$.

Indicator set. The framework accommodates any set of indicators. Table 2 lists the indicators available in our Chicago data, organized by category. The original four IMU components are retained for comparability with HRSA, supplemented by factors the 1970s formula omits.

Table 2: Candidate indicators for the underservice score.

Category	Indicator	Source	Level
Access	CTA transit time to nearest FQHC	CTA/OTP	Block
Access	Provider-to-population ratio	NPPES+Census	Tract
Coverage	Uninsured rate	Chicago Health Atlas	Comm. area
Burden	Diabetes prevalence	Chicago Health Atlas	Comm. area
Burden	Hypertension prevalence	Chicago Health Atlas	Comm. area
Demog.	Poverty rate	ACS 5-Year	Tract
Demog.	% aged 65+	Decennial	Block
Demog.	Infant mortality rate	Chicago Health Atlas	Comm. area

Normalization. We use two normalization functions depending on the indicator type.

For indicators with a natural policy threshold (e.g., 30 minutes of transit time), we use a sigmoid:

$$\varphi_k(x) = \frac{1}{1 + e^{-\kappa_k(x - \mu_k)}}, \quad (14)$$

where μ_k is the threshold (e.g., $\mu = 30$ for transit minutes) and κ_k controls the steepness. This produces a smooth inflection at the policy-relevant boundary.

For all other indicators we use the percentile rank:

$$\varphi_k(x) = \frac{\text{rank}(x)}{N}, \quad (15)$$

where N is the number of tracts. This is robust to outliers and directly interpretable: a tract scoring 0.85 is in the 85th percentile of need on that indicator. The CDC Social Vulnerability Index uses an analogous approach [2].

Weighting. Equal weights ($w_k = 1/K$) serve as the default. The framework supports user-adjustable weights, allowing the decision maker to specify which factors matter and by how much. This separates the policy question (what should count as underserved?) from the optimization question (given a definition, where should facilities go?). The model finds the best configuration for any weight profile.

Dynamic and static components. Among the indicators in Table 2, transit time to the nearest FQHC is the only component that changes when a facility is added or removed. All other indicators are fixed characteristics of the tract. This means the score responds to the decision variable: when FalCom places a facility in a district, the transit component of $S(t)$ decreases for nearby tracts, lowering the district energy. The optimizer can observe how each candidate location improves the system-wide score.

3.3.6 Probabilistic Extension

Two of the four original IMU indicators come from the American Community Survey, where in approximately 50% of census tracts the margin of error exceeds 50% of the estimate [13]. Rather than computing a single deterministic score, we model each ACS-sourced indicator with its reported uncertainty distribution and produce a probability of underservice for each tract.

For each tract t , let $\hat{x}_k(t)$ and $\sigma_k(t)$ denote the ACS point estimate and standard error for indicator k . We draw M samples

$$x_k^{(m)}(t) \sim \mathcal{N}(\hat{x}_k(t), \sigma_k(t)^2), \quad m = 1, \dots, M, \quad (16)$$

compute the score $S^{(m)}(t)$ for each sample, and report

$$P(\text{underserved} \mid t) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}[S^{(m)}(t) > \tau], \quad (17)$$

where τ is any threshold of interest. A tract with a point estimate just above τ but high variance may have only a 40% probability of truly exceeding it. This replaces the hard cutoff with a continuous risk measure.

For the FalCom energy function, the expected score $\mathbb{E}[S(t)]$ can be used directly, or a risk-averse variant $\mathbb{E}[S(t)] + \lambda\sqrt{\text{Var}[S(t)]}$ that penalizes uncertainty, directing resources toward tracts where either need is confidently high or measurement uncertainty is itself a problem warranting investment in better data.

3.3.7 District-Level Aggregation

FalCom optimizes over districts, not individual tracts. The energy of a district d with assigned facility j is the population-weighted average of its tracts' scores, modulated by the travel time to the facility:

$$E(d) = \frac{1}{\text{Pop}(d)} \sum_{t \in d} \text{Pop}(t) \cdot S(t) \cdot g(\tau(t, j)), \quad (18)$$

where $\tau(t, j)$ is the CTA transit time from tract t to facility j and g is a monotone increasing function (e.g., $g(x) = 1 + x/30$) so that tracts far from their assigned facility contribute more energy. This combines need (S) with access (g) in a single objective that FalCom minimizes across the system.

3.3.8 Policy Design: Continuous Funding Allocation

Any improved scoring methodology faces the political constraint described in Section 3.3.3: existing FQHCs will oppose any formula change that threatens their funding. We propose a transition mechanism that evaluates all sites under the new score without defunding any existing clinic.

Freeze the floor, redirect the growth. In the transition year, every existing FQHC receives a funding floor equal to its current grant amount. The continuous score is computed for all districts. In subsequent years, the floor remains nominally frozen: no site receives less than it received the previous year. All new appropriations above the aggregate floor are allocated proportionally to the district scores. Sites in the highest-need districts receive the growth; sites in lower-need districts receive flat nominal funding.

Federal FQHC appropriations have grown from \$554 million in 1990 to over \$6 billion today. Inflation alone erodes the frozen floor by 2–3% per year in real terms. Over a 10–15 year horizon, a site whose district has genuinely improved receives the same nominal amount but substantially less in real dollars, while high-need districts grow. No clinic is defunded. No designation is revoked. The growth is redirected.

Annual review cycle. Each year, the partition is recomputed with updated data (new providers, population shifts, new or closed sites). The score provides HRSA with two outputs:

1. Where the next new facility should go (the district with the highest energy after optimization).
2. Which existing districts have improved enough that their growth funding can be reallocated (gradual rebalancing).

This makes the scoring methodology reformable. Because no single annual update can cause a funding cliff, the formula can be improved incrementally as better data or indicators become available.

3.4 Referral System Design

TODO: Each FQHC must be within a certain CTA travel time of a referral hospital. Referral constraints. Integration with the hierarchical model.

4 Experiments

4.1 Data Sources

Census and demographic data. The newest block-level data is from the 2020 Decennial Census (DHC). Block-level demographic data is only released with the decennial census (every 10 years). Poverty rate data comes from ACS 5-Year 2020 (mean 19.0% at tract level). Total Chicago population: 2,783,891; population 65+: 356,716.

Provider data. We use NPES Data Dissemination V.2 (February 2026), which contains 77,590 Chicago providers from 9.3 million total records nationwide. This includes all provider types; we filter for primary care physicians.

FQHC demand and capacity (UDS). The Uniform Data System (UDS) collects annual reports from every HRSA-funded health center containing detailed information about patients served, medical/dental/behavioral visits, demographics, workforce (physicians, nurse practitioners), services, and finance. This dataset provides facility capacity, actual patient demand, service utilization, and staffing constraints. Almost no operations research papers use this data.

Area Health Resource File (AHRF). This dataset covers every U.S. county and includes hundreds of variables: physician counts, hospital beds, demographics, poverty rates, infant mortality, insurance coverage, and workforce data.

Table 3: Chicago geographic and provider data files.

File	Level	Year	Contents
chicago_blocks_enriched.pkl	Block	2020	Population, age 65+, poverty rate
chicago_block_age65.csv	Block	2020 DHC	GEOID20, total population, population 65+
chicago_tract_poverty.csv	Tract	2020 ACS	Poverty rate, population below poverty
chicago_providers.csv	Provider	2026 NPES	77,590 Chicago NPI providers

4.2 Implementation

[TODO: FalCom configuration for Chicago.](#) [CTA travel time computation.](#) [Dashboard description.](#)

4.3 Results

[TODO](#)

5 Discussion

[TODO](#)

6 Conclusion

[TODO](#)

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